



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT317

Final Examination

Date: 9/17/2025

Total Time: 2.00 Hours

Course Title: Mathematical Method

Semester: Spring 2025

Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

- | | CLOs | Marks |
|---|------|-------|
| Q1. (a) Prove that the generating function of the Bessel function $J_n(x)$:
$e^{\frac{1}{2}x(t - \frac{1}{t})} = \sum_{-\infty}^{\infty} t^n J_n(x).$ | CLO2 | [4] |
| (b) Prove the following recurrence formulae for the Bessel function $J_n(x)$:
(i) $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x).$
(ii) $J_n(x) = \frac{x}{2n} [J_{n-1}(x) + J_{n+1}(x)].$ | CLO3 | [6] |
| Q2. (a) Define Bessel's differential equation, and solve Bessel's differential equation and function. | CLO1 | [10] |
| Q3. (a) Prove the following trigonometric expansion involving the Bessel function $J_n(x)$:
(i) $\cos(x \sin \varphi) = J_0(x) + 2[\cos 2\varphi J_2(x) + \cos 4\varphi J_4(x) + \dots].$
(ii) $\sin(x \cos \varphi) = 2[\cos \varphi J_1(x) - \cos 3\varphi J_3(x) + \dots].$ | CLO2 | [7] |
| (b) Prove that $J_2(x) = J_0''(x) - \frac{J_0'(x)}{x}.$ | CLO2 | [3] |
| Q4. (a) Define the Hermit differential equation. | CLO1 | [1] |
| (b) Deduce the generating function for Hermit polynomial. | CLO2 | [5] |
| (c) Prove that $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}).$ | CLO3 | [4] |
| Q5. (a) Define the Laguerre's differential equation. | CLO1 | [1] |
| (b) Prove that: (i) $L_n'(0) = -n$, and (ii) $L_n''(0) = \frac{n(n+1)}{2}.$ | CLO3 | [4] |
| (c) Using the definition of Laguerre polynomial find the values of $L_0(x), L_3(x), L_5(x).$ | CLO3 | [5] |
| Q6. (a) Define the Legendre's differential equation and solve it. | CLO1 | [7] |
| (b) Using the definition of Legendre polynomial find the values of $P_2(x), L_5(x), L_7(x).$ | CLO2 | [3] |



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE321: Theory of Computing

Batch: 24th
 Date: 20/09/2025

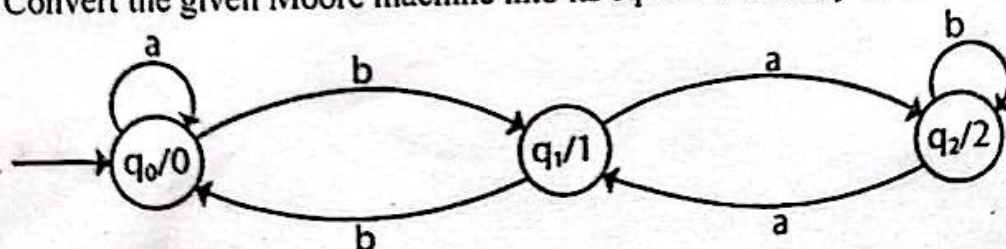
Final Examination

Semester: Spring 2025
 Time: 2 Hours
 Total Marks: 40

Answer any five questions.

Marks: 5×8 = 40

- | | | Marks | CLOs |
|----|--|-------|------|
| 1. | (a) Define Mealy machine with mentioning its possible tuples. | 2 | CLO2 |
| | (b) Design a Mealy machine for a binary input sequence such that if it has a substring 111, the machine outputs X, if the input has substring 100, it outputs Y, otherwise it outputs Z. | 6 | CLO3 |
| 2. | (a) Define the ambiguous situation of a grammar. | 2 | CLO3 |
| | (b) Design a Moore machine with the input alphabet {0, 1} and output alphabet {Y, N}, which produces Y as output if input sequence contains 11101 as a substring otherwise, it produces N as output. | 6 | CLO3 |
| 3. | (a) Illustrate parse tree. | 2 | CLO2 |
| | (b) Derive the string "aabbabba" for leftmost and rightmost derivation using a CFG given by,
$S \rightarrow aB \mid bA$
$A \rightarrow a \mid aS \mid bAA$
$B \rightarrow b \mid bS \mid aBB$ | 3×2=6 | CLO3 |
| 4. | (a) Write a short note on Context Free Grammar (CFG). | 2 | CLO2 |
| | (b) For the grammars $G: E \rightarrow E+E \mid E-E \mid id$, considering string "id+id-id" show that G is ambiguous. | 6 | CLO3 |
| 5. | (a) Derive production rules for the regular expression $(a+b)^*$. | 3 | CLO3 |
| | (b) Convert the given Moore machine into its equivalent Mealy machine. | 5 | CLO3 |

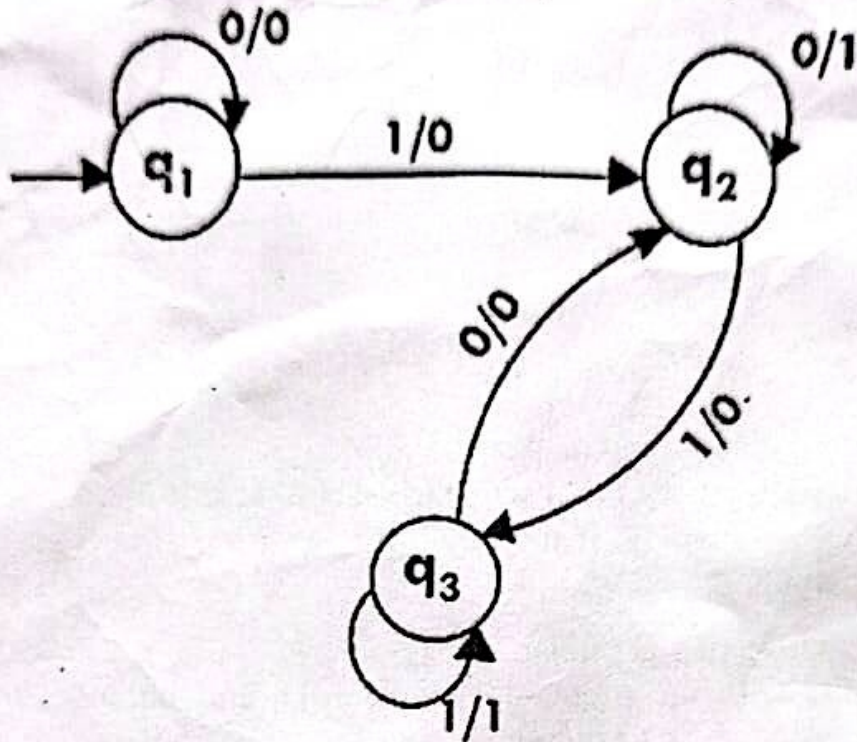


4×2=8 CLO3

6. Suppose,
 $S \rightarrow (L) \mid a$
 $L \rightarrow L, S \mid b \mid s$
 Find leftmost derivation for
 (i) $(a, (a, a))$
 (ii) $(a, ((a, a), s))$



7. Convert the following Mealy machine into equivalent Moore machine:





Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313: Database Management System

Batch: 24th

Date: 07/09/2025

Final Examination

Semester: Spring 2025

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings.

5x8= 40

1. (a) State the guidelines for writing SQL Statements. CLO 2 3
 (b) What is meant by a group function? Write the SQL to show the max, min, sum and average salary distinctly from the employees table. CLO 4 5

2. (a) State the comparison conditions used in SQL statements. CLO 2 2
 (b) Write the SQL and relational algebra to determine different types of outer-join for the following relations: CLO 4 6

People			Menu	
Name	Age	meal	meal	Day
Bill	22	Hamburger	Hamburger	Monday
Sally	28	Pizza	Pizza	Tuesday
Marry	23	Soft-drinks	Chicken	Wednesday
Tony	32	Shrimp	Pasta	Thursday
			Tacos	Friday

3. (a) Write the SQL to show employee_id, last_name, salary, manager_id from the employees table whose manager_id not consisting 100, 101 and 201. CLO 4 2
 (b) Consider the following Employees table: CLO 4 6

Emp_id	First_name	Last_name	Email	Job_id	Salary	Department_id
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- i) Update employee 114's job_id and salary to match that of employee 205 using subquery.
- ii) Insert the following data into the above employees table using subquery:

Emp_id	Last_name	Email	Job_id	Salary	Department_id
99999	Taylor	123@gmail.com	St_clerk	50000	50

4. (a) Using the Employees table of 'Q3(b)', Write the SQL to show the last_name, job_id and salary whose job_id containing 141 employee_id not only but also greater than the 143 employee_id's salary. CLO 4 4

- (b) Show the differences between join and natural join in DBMS with example. CLO 4 4

5. (a) State the logical conditions used in SQL statements. CLO 2 2
 (b) Write the SQL to show employee_id, last_name, job_id, salary from the employees table whose salary not only more 10000, but also sr_officer as a job_id. CLO 4 3
 (c) Write the SQL to show last_name, job_id, salary from the employees table whose job_id not only either 'SA_REP' or 'AD_PRES', but also earned more than 15000 as a salary. CLO 4 3

6. (a) State the rules of operator precedence used in SQL statements. CLO 2 2
 (b) Explain the different parts of SQL. CLO 2 3

(c) State the guidelines for creating an Index in DBMS.

CLO 2 . 3

7. (a) Explain Normalization with an example.

CLO 4 4

(b) State the similarities and dissimilarities between 3NF and BCNF.

CLO 4 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Batch: 24th

Semester: Spring 2025

Date: 10/09/2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer five questions of the following.

5 × 8 = 40

1(a). State the important operational characteristics (/notes) regarding the Distance Vector Routing Algorithm.

CLOs
CLO 5 Marks
2

1(b). Consider a network consisting of three routers: R1, R2, and R3. The links connecting these routers have the following weights, which can represent distances, costs, or delays:

CLO
4,5 6

- R1-R2: Weight = 4.

- R1-R3: Weight = 7

- R2-R3: Weight = 2

Draw the given network and calculate the best path for the given network using **Distance Vector Routing Algorithm**.

2(a). Briefly explain the fundamental difference in how **Link-State** and **Distance Vector** routing algorithms gather information about the network topology.

CLO 5 2

2(b). Consider a network topology consisting of five routers: A, B, C, D, and E. The links connecting these routers have the following associated costs:

CLO 4 6

- A-B: Cost = 3

- A-C: Cost = 2

- B-D: Cost = 4

- C-D: Cost = 1

- C-E: Cost = 5

- D-E: Cost = 2

Assume that Router A is the "source" node that needs to compute its least-cost paths to all other nodes in the network using a Link-State Routing Protocol. Apply **Dijkstra's Algorithm** to compute the least-cost paths from Router A to all other routers and determine Router A's forwarding table.

3(a). Define Public and Private network address.

CLO 2 2

3(b). "InnovateTech Solutions" is a growing company with an internal network connecting 150 employee computers. For their internal network, they use private IP addresses from the 192.168.10.0/24

CLO 2 6

range. Their corporate router, named IT-Gateway, acts as the sole connection point between their private network and the public Internet. The IT-Gateway router has been assigned one single public IP address: 203.0.113.50 by their Internet Service Provider. All employee traffic going to the Internet must pass through this router.

Additionally, "InnovateTech Solutions" hosts a critical internal server, DataVault, which has the private IP address 192.168.10.25. This DataVault server needs to be accessible from specific trusted partner networks on the Internet via a dedicated public IP address, 203.0.113.60.

Explain why "InnovateTech Solutions" chose to use the 192.168.10.0/24 range for its internal network, and what the primary purpose of implementing NAT on the IT-Gateway router is in this scenario?

- | | | | |
|-------|--|-------|---|
| 4(a). | Write about the components of Simple Mail Transfer Protocol (SMTP). | CLO 4 | 2 |
| 4(b). | Explain the fundamental role of the SMTP in email communication, detailing its operational mechanism from a user sending an email to its delivery in the recipient's mailbox. | CLO 4 | 6 |
| 5(a). | Write the features of Open Shortest Path First (OSPF) protocol. | CLO 4 | 2 |
| 5(b). | Discuss the "Count to Infinity Problem" in Distance Vector Routing, elaborating on its fundamental cause and the circumstances under which it typically arises. Furthermore, describe the two primary solutions employed to mitigate its occurrence. | CLO 5 | 6 |
| 6(a). | Define Name server and Zone of Domain Name Server (DNS). | CLO 5 | 2 |
| 6(b). | "GlobalLink Enterprises" is a multinational technology company that operates various online services. Their primary public-facing website is www.global-enterprises.com . They also manage an internal intranet, intranet.global-enterprises.local , which is exclusively accessible to employees within their corporate network. Recently, GlobalLink expanded its operations into Canada, launching a new regional service accessible via canada.global-enterprises.ca . Analyze the DNS infrastructure and operations of "GlobalLink Enterprises" as well as corresponding IP address mappings based on the scenario provided. | CLO 5 | 6 |
| 7(a). | What is Dynamic Host Configuration Protocol (DHCP)? | CLO 4 | 2 |
| 7(b). | Explain the messages of DHCP protocol. | CLO 4 | 6 |



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315: Microprocessor & Embedded System

Batch: 24th

Date: 15/09/2025

Final Examination

Semester: Spring 2025

Time: 2 Hour

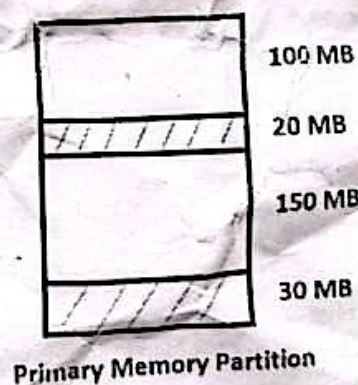
Total Marks: 40

Answer any five questions of the following.

$5 \times 8 = 40$

1. (a) What is a Microcontroller? Explain about Dual CPU Core Chip CLO3 4
- (b) Distinguish between Intel Core i7 and Intel Core i5. CLO3 4
2. (a) Explain about Processor Technology and its Classification. CLO3 4
- (b) Draw the architecture of Embedded System Hardware for Home Control Automation like Smart Home and briefly explain the key parts. CLO3 4
3. (a) What are Contiguous and Non-Contiguous Memory Allocation? CLO3 3
- (b) Apply First-fit, Next-fit, Best-fit, Worst-fit for the following example and Justify which Algorithms is better for variable size partitioning. CLO3 5

Process ID	Size
P1	10 MB
P2	15 MB
P3	20 MB
P4	7 MB
P5	8 MB



4. (a) Mention Types of Interrupt in 8086. CLO2 2
- (b) Explain Interrupt Execution with the help of Interrupt vector table in the 8086 microprocessor. CLO3 3
- (c) Briefly Mention Hardware Interrupt and Interrupt Priority in 8086. 3
5. (a) Draw the Pin Diagram of Microprocessor 8086 and Explain the followings: CLO1 4
 - i. Address Status Line
 - ii. NMI
- (b) Briefly explain the followings: CLO3 4
 - i. Embedded Processor
 - ii. ASSP

- iii. Application Specific Processor
- iv. Digital Signal Processor

6.

- (a) Draw the diagram for a Micro Program Control Unit and Explain the Instruction Cycle for [MOV X₁, X₂] CLO1
- (b) Analyze Wilkes Design for a Micro Programmed CU for (001) and (011) input. CLO3

7.

- (a) Answer the following short Questions: CLO2
 - i. What you need to do for Loading data segment address?
 - ii. What is the function of .CODE Segment on assembly program?
 - iii. Write one instruction to clear the contents of register AX.

- (b) Write an 8086-assembly language program that: CLO2
 - Takes three input numbers from the user.(Show input messages)
 - Adds them together.
 - Displays the final sum.